# 3.1 The Chain Rule

Name _____ Period ____ Date _____

AP Calculus 3.1 · Student sheet

**I can…**

- identify the outer and inner functions of a composite function.
- use the Chain Rule, including the Power Rule for functions.
- use the Chain Rule with $e^u$, trig functions, and tables of values.

**Vocabulary** · fill in the missing word

**Composite function** — $(f \circ g)(x) = f(g(x))$: the _____ of g is the input of f

**Chain Rule** — $\frac{d}{dx} f(g(x)) =$ _____

**Power Rule for functions** — $\frac{d}{dx} [u(x)]^n =$ _____

## Part 1 · Expand or not?

$f(x) = (2x + 1)^2$

| Method | f′(x) | Matches? |
|---|---|---|
| Expand: $4x^2 + 4x + 1$ | | |
| Guess: $2(2x + 1)$ | | |
| Fixed guess | | |

## Part 2 · Notes and examples (fill in each note, then work the example)

**Chain Rule:** if $y = f(u)$ and $u = g(x)$, then $\frac{dy}{dx} = \frac{dy}{du} \cdot$ _____.

### Example 1

$y = u^5$ and $u = x^2 + 1$. Find $\frac{dy}{dx}$.

**Chain Rule (function form):** $\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot$ _____.

### Example 2

$f(2) = 5$, $f'(2) = 3$, $g(1) = 2$, $g'(1) = -4$, $f'(5) = 6$. $h(x) = f(g(x))$. Find $h'(1)$.

**Power Rule for functions:** $\frac{d}{dx} [u(x)]^n = n[u(x)]^{n-1} \cdot$ _____.

### Example 3

Find $f'(x)$ for $f(x) = (3x^2 - 5)^4$, then find $f'(1)$.

**Exponential and trig:** $\frac{d}{dx} e^u = e^u \cdot u'$, $\frac{d}{dx} \sin u =$ _____ $\cdot u'$.

### Example 4

Find each derivative.

a. $y = e^{(x^2 - 3x)}$

b. $y = \sin(4x)$

### Example 5

Find the tangent line to $y = (x^2 - 3)^3$ at $x = 2$.

## Part 3 · Partner practice

### Partner 1

$f(x) = \cos(3x)$. Find $f'(\frac{\pi}{6})$.

### Partner 2

(AP® Practice) If $f(x) = (x^2 - 3)^4$, then $f'(2) =$

(A) 4 (B) 16 (C) 32 (D) 64